

Xavier Malara

Ottawa, ON | Bilingual (English/French) | xmala086@uottawa.ca | linkedin.com/in/xavier-malara | github.com/revaix7

EDUCATION

University of Ottawa

BASc Software Engineering (Co-op), Second Year

Ottawa, ON

Expected December 2029

EXPERIENCE

Engineering Outreach Instructor – STEM Camps & Cyber Range

Summers 2024 – 2026

University of Ottawa (Co-op work term: Summer 2026)

Ottawa, ON

- Taught **OSINT**, **hashing**, and **cybersecurity** fundamentals to high school classes of **30+** at the uOttawa Cyber Range, running each class through a hands-on scenario in a virtual environment
- Co-led a smart-home hacking workshop for **70+ students**, guiding them to exploit vulnerabilities in physical, Arduino-powered model houses and see real-world IoT security risks firsthand
- Authored and tested **100+** activity guides for Micro:bit, TinkerCAD, Sphero, MakeCode, and KnowBe4 security
- Designed and ran hands-on **Python**, **3D modeling**, **Micro:bit**, and **Arduino** lessons for groups of **30+ campers**, who left with their own working builds, 3D-printed designs, and Python programs
- Delivered a **week-long outreach program** in Sudbury, teaching Micro:bit coding (MakeCode), Sphero robotics, TinkerCAD 3D modeling, and hands-on science to **10+ Indigenous youth**

PROJECTS

OttaTransit | React, Vite, MapLibre GL, Node.js, Express, TanStack Query

[GitHub](#)

- Built a **live OC Transpo tracker** that maps nearby stops and shows real-time arrival countdowns for each route
- Wrote an **Express proxy** to decode GTFS-Realtime protobuf feeds, sidestepping CORS and hiding the API key
- Served live data from an in-memory snapshot refreshed every 15 seconds, so requests arriving mid-refresh never returned an empty map
- Animated vehicles between feed updates by dead reckoning from reported speed and bearing, keeping map movement smooth

TipRoute | React, React Router, Vite

[GitHub](#)

- Created a **trip planner** that estimates time, budget, and difficulty for multi-country, tip-to-tip routes
- Implemented a **dual-mode engine** that estimates the day range for a given route, or the reachable distance and checkpoint for a set number of days

Property Booking Platform | JavaScript, HTML/CSS, Supabase

[GitHub](#)

- Developed a French-language **booking platform** with user accounts, checkout flows, and auto-generated receipts
- Integrated **Supabase** for account and booking storage; converted images to AVIF/WebP to cut page weight

TECHNICAL SKILLS

Languages: Java, Python, C#, JavaScript, HTML/CSS

Frameworks & Libraries: React, React Router, Express, TanStack Query, MapLibre GL, Windows Forms

Developer Tools: Node.js, Vite, Git, GitHub, Supabase

Security: OSINT, hashing, cybersecurity fundamentals, smart-home/IoT security basics

Hardware & Fabrication: Arduino, Micro:bit, 3D modeling (TinkerCAD) & printing, basic welding

CERTIFICATIONS & TRAINING

Standard First Aid & Bronze Cross (Lifesaving Society) | WHMIS | Introductory welding training

VOLUNTEER EXPERIENCE

Community Volunteer | Soup Kitchen & School Events

Ottawa, ON

- Prepared and served meals at a community centre soup kitchen and organized incoming donations
- Coordinated with staff and fellow student volunteers to plan, set up, and run school events